Band-1 Euler / Boussinesq Cousin Blowup: Exact Statement Lock (Collapse 1.20)

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Document class. Literature + exact-sentence lock (Band-1 hygiene).
Collapse item. 1.20 — Euler / Boussinesq cousin blowup increments.
Sibling. Forced-NS Fefferman C/D audit = **H-1.19** (`2026-09-11_Forced_NS_Statement_Lock_Hygiene_Note`).
Non-negotiable. This note does **not** stamp Clay-closed, does **not** claim NS solved, does **not** claim ToE finished. **GATE-04** is locked out of scope. No invented numerical blowup certificates.

Abstract

September 2026 press (including *New Scientist*, 8 Sep 2026) frames a human+AI heat wave around Euler / Boussinesq / IPM blowup cousins and Navier–Stokes announcements. This Band-1 probe **locks exact PDE sentences**: what is already proved in the literature (forced vs unforced; smooth vs $C^{1,\alpha}$; with boundary vs without), and what the house keeps **OPEN**. Forced smooth-force blowups (Alpöge–Buckmaster, building on Córdoba–Martínez-Zoroa) are **different theorems** from unforced C^∞ 3D Euler on $\mathbb{R}^3/\mathbb{T}^3$, and both are **different** from Clay unforced Navier–Stokes (**H-4.12**). Cousin increments raise Band-1 priority; they do not close Band 4.

Keywords. 3D Euler; 2D Boussinesq; IPM; forced vs unforced; Elgindi; Chen–Hou; Buckmaster–Vicol; Alpöge–Buckmaster; statement lock; no Clay stamp.

1. House rules for this probe

1. **Statement lock before press.** The quantified PDE sentence is the claim; headlines are heat.
2. **Force-class in every sentence.** $f \equiv 0$ versus smooth force f changes the theorem.
3. **Regularity and domain in every sentence.** $C^{1,\alpha} \neq C^\infty$; cylinder-with-boundary $\neq \mathbb{R}^3$.
4. **Separate from Clay NS.** Unforced NS (**H-4.12**) stays **OPEN** here; see H-1.19 for Fefferman (A)–(D) hygiene.
5. **GATE-04 LOCKED.** Out of scope.
6. **No invented certificates.** Lit + lock only; no fake ODE blowup plots.

2. Exact sentences — PROVED (literature)

L1 — Elgindi (Annals of Mathematics, 2021)

PDE. Unforced ($f \equiv 0$) 3D incompressible Euler on $\mathbb{R}^3$.
Data. Axisymmetric without swirl; velocity in $C^{1,\alpha}$ (finite energy after localization).
Claim (proved). There exist such solutions that develop a singularity in finite time.
Not this theorem. C^∞ blowup; general (non-axisymmetric) smooth Euler; Clay NS.

L2 — Chen–Hou (arXiv 2022/2023; *Multiscale Model. Simul.* 2025; PNAS survey 2025)

PDE. Unforced 2D Boussinesq; unforced 3D axisymmetric Euler.
Data / domain. Smooth finite-energy initial data **with boundary** (cylinder / smooth bounded domain; Hou–Luo scenario).
Claim (proved, computer-assisted). Nearly self-similar finite-time blowup, stable in stated weighted norms.
Not this theorem. Boundary-free C^∞ Euler on $\mathbb{R}^3/\mathbb{T}^3$; Clay NS. PNAS framing itself: the *general* singularity problem remains unresolved.

L3 — Alpöge–Buckmaster (public ~2026-09-07; Lean on released items)

PDE. Incompressible porous medium (IPM); 2D Boussinesq; **3D incompressible Euler**.
Force. Smooth space-time external force (force remains smooth through the singular time).
Claim (proved-as-released). Finite-time singularity from smooth data with smooth force; Córdoba–Martínez-Zoroa forced-blowup program pushed to smooth force and to Euler; Lean formalizations accompany released items.
Not this theorem. Same conclusions with $f \equiv 0$; Clay unforced NS (**H-4.12**).
Credit line (authors). Ideas of Córdoba–Martínez-Zoroa; LLM-assisted completion (Buckmaster public statement).

L4 — Buckmaster–Vicol (Annals of Mathematics, 2019)

PDE. 3D Navier–Stokes.
Claim (proved). Weak solutions with finite kinetic energy are **not unique** in that weak class (convex integration); Hölder dissipative Euler can arise as strong vanishing-viscosity limits of such weak NS solutions.
Not this theorem. Non-uniqueness of Leray–Hopf solutions; classical smooth blowup; Clay gate.

3. Exact sentences — OPEN (house locks)

O1 — Unforced C^∞ 3D Euler (house H-4.13)

Whether smooth, finite-energy solutions of the **unforced** ($f \equiv 0$) 3D incompressible Euler equations on $\mathbb{R}^3$ or $\mathbb{T}^3$, **without** boundary or low-regularity crutches, can blow up in finite time remains **OPEN** for this house.
September 2026 press claims that frame an "unforced Euler resolved" announcement are recorded as **heat only** and are **not** stamped proved here.

O2 — Clay unforced Navier–Stokes (house H-4.12 / CLAY-NS)

Whether smooth solutions of **unforced** ($f \equiv 0$) 3D incompressible Navier–Stokes exist globally for all admissible smooth data, or can develop a finite-time singularity, remains **OPEN** until the Clay Mathematics Institute moves.
Forced Fefferman (C)/(D) heat and cousin Euler/Boussinesq increments are **different theorems**.
This note does not close Clay.

O3 — Cousin $\neq$ prize PDE

IPM / Boussinesq / forced-Euler increments are Band-1 **cousins**. They are not Millennium certificates and not a ToE finish line.

4. Forced vs unforced — one table

Result	Force	Regularity	Domain / symmetry	House status
Elgindi 2021	$f = 0$	$C^{1,\alpha}$	$\mathbb{R}^3$, axisym. no swirl	Proved lit; not C^∞ open Euler
Chen–Hou 2022–25	$f = 0$	smooth	**with boundary**, axisym.	Proved (CAS); not boundary-free open Euler
Alpöge–Buckmaster 2026	**smooth** f	smooth	IPM / Boussinesq / 3D Euler	Forced cousin **proved-as-released**; not unforced
Buckmaster–Vicol 2019	weak NS	weak / low	$\mathbb{T}^3$	Weak non-uniqueness; not Clay smooth
Unforced C^∞ Euler	$f = 0$	C^∞	$\mathbb{R}^3/\mathbb{T}^3$, no crutches	**OPEN (H-4.13)**
Unforced Clay NS	$f = 0$	smooth	$\mathbb{R}^3/\mathbb{T}^3$	**OPEN (H-4.12)** — no stamp

5. 2026 heat (verified mention; not a proof stamp)

New Scientist (8 Sep 2026; follow-up 9 Sep) and concurrent coverage describe a human+AI flurry: Buckmaster–Alpöge releasing smooth-force Euler / Boussinesq / IPM blowups, and OpenAI announcing Navier–Stokes / Euler-related formalizations amid a credit dispute. That chronology is **accepted as heat**. It is **rejected** as automatic house closure of Clay NS, unforced C^∞ Euler, or ToE.

Tao (blog, 7 Sep 2026): Alpöge–Buckmaster forced cousins are breakthroughs on the Córdoba–Martínez-Zoroa line; unforced goals remain the longer program.

6. Cross-references

ID	Title	Relation
H-1.19	Forced-NS statement lock (Fefferman C/D)	Sibling hygiene; NS force audit
H-1.20	This note	Cousin Euler / Boussinesq lock
H-2.40	Unforced 3d Euler under extra symmetry	Partial lit progress only
H-4.12	Unforced Clay NS	**OPEN**
H-4.13	Unforced 3d Euler, no symmetry	**OPEN**

7. Lab paths

- Lab: `/workspace/mission-control/experiments/band1/euler_boussinesq_cousin/`
- Ledger: `literature_ledger.json`
- Sentences: `STATEMENT_LOCK.md`
- Paper: `papers/2026-09-11_Band1_Euler_Boussinesq_cousin_statement_lock.md` (+ PDF)

8. Mandatory honesty line

Clay Navier–Stokes is not closed in this house. Unforced C^∞ 3D Euler without crutches is not closed in this house. Forced cousin blowups are real increments and different theorems. GATE-04 stays locked. ToE is not finished.